

Memory File — Ritik Ghoghari Thesis Session (Sep 15, 2026)

Paste this into Claude when starting a new conversation

Who I Am

- Name: Ritik Ghoghari
 - Degree: M.Sc. Data Science, AI & Digital Business — GISMA University of Applied Sciences, Berlin
 - Supervisor: Professor Mohamad Hoseini
 - Deadline: 26 September 2026
 - Working directory: C:\Users\Allah o akbar\thesis\
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Thesis: Catching Greenwashing from Space

Automated pipeline that:

1. Reads palm oil company sustainability reports (PDF)
 2. Extracts deforestation claims using NLP models
 3. Measures actual forest loss via satellite (Google Earth Engine)
 4. Combines both into a mismatch score 0–100
 5. High score = company claims zero deforestation but satellites show clearing = greenwashing signal
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Verified Numbers — Never Change Without Re-Running

- 11 companies, 290 mills, 890,168 ha post-2020 forest loss
 - GAR: 50 mills | 171,236 ha | 13.87%
 - SD Guthrie: 42 mills | 142,131 ha | 14.41%
 - Wilmar: 45 mills | 127,606 ha | 11.40%
 - KLK: 30 mills | 92,209 ha | 12.18%
 - Astra Agro: 34 mills | 87,198 ha | 10.76%
 - IOI: 15 mills | 74,385 ha | 19.04% (highest intensity)
 - Musim Mas: 18 mills | 53,060 ha | 11.45%
 - Genting: 15 mills | 44,362 ha | 11.61%
 - First Resources: 16 mills | 40,090 ha | 10.29%
 - Bumitama: 14 mills | 30,549 ha | 8.66%
 - SIPEF: 11 mills | 27,341 ha | 10.42%
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Mismatch Score Ranking — Verified

1. KLK — 65.6
2. GAR — 63.1
3. Musim Mas — 56.2

4. IOI — 56.1
5. SD Guthrie — 55.2
6. Wilmar — 50.3
7. First Resources — 41.7
8. Genting — 39.5
9. Bumitama — 38.5
10. SIPEF — 26.1
11. Astra Agro — 17.6

Scoring Formula (CANONICAL — professor approved)

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Mismatch Score = 0.35 × Forest Loss
                 + 0.35 × Specificity
                 + 0.15 × Sentiment
                 + 0.15 × Spatial Match
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Weights sum to 1.0. Never use 0.30/0.20 variant — that is stale.

NLP Models (5 total)

1. climatebert/environmental-claims — claim detection (F1 0.981)
 2. climatebert/distilroberta-base-climate-specificity — specificity scoring
 3. ProsusAI/finbert — sentiment (feeds mismatch score, weight 0.15)
 4. climatebert/distilroberta-base-climate-sentiment — comparison signal only
 5. nbroad/ESG-BERT — ESG category per claim
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Satellite Stack (4 total)

1. Hansen GFC v1.13 via Google Earth Engine — primary loss (30m, 99.5% accuracy)
 2. Descals oil palm map — 56% of cleared land = oil palm; IOI 89.8%, Bumitama 16.6%
 3. RADD SAR alerts — radar cross-check, confirmed 4 of 5 top mills
 4. Prithvi-EO-2.0-300M — foundation model, linear probe $p=0.805$, IoU=0.765
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Key Files

- main.ipynb — all analysis
- scripts/build_scores_v2.py — canonical scoring (23 tests pass)
- Results/master_scores.csv — final mismatch scores
- Results/satellite_images/clearing_maps/ — 290 Sentinel-2 clearing maps at 300 DPI
- latex/thesis_latex.zip — Overleaf upload ready
- data/mills/master_forest_loss.csv — 11 companies, 290 mills verified
- docs/MEETING_5_SPEAKING_NOTES.md — full supervisor meeting script
- docs/WEBSITE_SPEAKING_NOTES.md — website walkthrough script

- latex/defense_speaking_notes.html — architecture diagram speaking notes
 - diagrams/thesis_architecture.html — pipeline architecture diagram
 - website/index.html — interactive showcase website
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Best Satellite Images to Show Supervisor

1. Results/satellite_images/clearing_maps/ → ioi_syarimo_clearing_map.png (12,318 ha — highest single mill)
 2. gar_nagasakti_clearing_map.png (10,197 ha — GAR worst mill, company claims 99.8% traceability)
 3. wilmar_sabahmas_clearing_map.png (clearing every year 2021–2025, unique continuous pattern)
 4. astraagro_gunungsejahtera_clearing_map.png (counter-example — big loss, low mismatch score)
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Architecture Diagram — 3 Lanes

- Top lane: 11 PDFs → ClimateBERT claim extractor → 55 claims → specificity + sentiment scoring
 - Bottom lane: 290 mill GPS → Google Earth Engine → Hansen 10km buffer → forest loss ha
 - Two lanes NEVER share data until final scoring engine
 - Scoring engine outputs: rankings CSV + website + validation (RADD, Prithvi, Descals)
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Buffer Sensitivity

- Tested at 5, 10, 15, 20, 30 km
 - 10km canonical (professor approved, matches PalmWatch methodology)
 - Adjacent radii stable ($p=0.80$ to 0.96); 10km vs 30km unstable ($p=0.49$)
 - NO 2km or 3km buffer images exist in this project
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What's 100% Done (as of Sep 8, 2026)

- 290 Sentinel-2 clearing maps at 300 DPI
- Prithvi evaluated across all 290 mills
- Gold-label NLP evaluation (150 rows)
- Interactive website (website/)
- All 9 LaTeX chapters (~14,100 words, 0 em dashes, 0 AI buzzwords)
- Buffer sensitivity 5–30km
- Descals oil palm overlay
- Overleaf zip ready (latex/thesis_latex.zip)
- 23 unit tests passing

Remaining Human Actions

1. Upload latex/thesis_latex.zip to Overleaf → compile PDF
 2. Send compiled PDF to Professor Mohamad Hoseini
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Key Limitation (Always Mention)

Forest loss near a mill does not prove causation. Other actors share the catchment area. This measures correspondence, not causation. Same limitation as PalmWatch — documented honestly in the thesis.

Writing Rules

- No em dashes
 - No AI vocabulary: leverage, robust, comprehensive, pivotal, holistic, seamless, delve
 - Every number cited with Hansen et al. 2013
 - Short sentences, human tone
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User Preferences

- Communicates in Hinglish (Hindi + English mix)
- Wants visual, supervisor-ready artifacts
- Caveman mode active (terse responses)
- Likes before/after satellite image comparisons for supervisor demos